ASSIGNMENT 3.1

Multi-Agent Coordination: Contract Net Protocol

Student	[Student Name]	Roll No	[Roll No]
Course	21CSE525T - Agent Technology	Project	Placement Preparation Agent

Unit 3 - Negotiation between one buyer agent and three self-interested mentor agents. File: assignments/a3_1/negotiation.py

1. Objective

Coordinate several self-interested agents to award one mock-interview contract, using a call for proposals, counter-proposals and a handshake.

2. Participants

- **StudentPrepAgent** (manager/buyer): budget Rs 12,000, needs 8 slots, scores offers with the buyer utility.
- **MentorAgent-A**: quality 9, reservation price Rs 5,200, flexibility 0.30.
- **MentorAgent-B**: quality 7, reservation price Rs 3,800, flexibility 0.45.
- **MentorAgent-C**: quality 8, reservation price Rs 4,600, flexibility 0.22.

3. Protocol

Phase 1 - Call for proposals: each mentor opens at an inflated fee (1.45 to 1.75 times its reservation price).

Phase 2 - Negotiation (up to 6 rounds): the buyer raises its target price by 6% and lowers its acceptance threshold by 0.035 each round. Each mentor concedes a fraction of the gap set by its flexibility, never below its reservation price, and withdraws when it cannot move.

Phase 3 - Handshake: the first bid whose buyer utility meets the threshold is awarded (ACCEPT-PROPOSAL and ACK). All other mentors receive REJECT-PROPOSAL.

4. Utility functions

Buyer: U = 0.45 x (1 - fee/budget) + 0.35 x (quality/10) + 0.20 x min(slots/needed, 1). Lower price, higher quality and more slots score higher.

Seller (mentor): margin over its reservation price (weight 0.7) plus slot efficiency (weight 0.3). Mentors are self-interested, so they never accept less than their reservation price.

5. Implementation

```
class MentorAgent:
    def __init__(self, name, quality, min_fee, max_slots, flexibility, seed):
        self.name = name
        self.quality = quality            # 0-10 training quality rating
        self.min_fee = min_fee            # reservation price (won't go below)
        self.fee = min_fee * random.Random(seed).uniform(1.45, 1.75)
        self.max_slots = max_slots
        self.slots = max_slots
        self.flexibility = flexibility    # concession rate per round
        self.rng = random.Random(seed)
        self.active = True

    def utility(self, fee, slots):
        """Seller utility: profit margin x slot efficiency."""
        margin = (fee - self.min_fee) / self.min_fee
        eff = slots / self.max_slots
        return round(margin * 0.7 + eff * 0.3, 4)

    def propose(self, rnd):
        if not self.active:
            return None
        return {"agent": self.name, "fee": round(self.fee, 2), "slots": self.slots,
```

```
                "quality": self.quality, "round": rnd,
                "utility": self.utility(self.fee, self.slots)}

    def counter(self, target_fee, rnd):
        """Modify stance: concede toward the manager's target, but never below cost."""
        gap = self.fee - target_fee
        new_fee = self.fee - gap * self.flexibility
        if new_fee &lt; self.min_fee:
            if self.fee &lt;= self.min_fee * 1.02:
                self.active = False
                log(self.name, "REFUSE", f"cannot go below reservation Rs.{self.min_fee}. WITHDRAWING.")
                return None
            new_fee = self.min_fee
        # occasionally add a slot as a sweetener
        if self.rng.random() &lt; 0.4 and self.slots &lt; self.max_slots + 2:
            self.slots += 1
        self.fee = new_fee
        log(self.name, "PROPOSE", f"round {rnd}: Rs.{self.fee:,.2f} | {self.slots} slots | "
                                  f"quality {self.quality}/10 | own-utility={self.utility(self.fee, self.slots)}")
        return self.propose(rnd)

class StudentPrepAgent:
    def __init__(self, budget, needed_slots, w_cost=0.45, w_qual=0.35, w_slot=0.20):
        self.name = "StudentPrepAgent"
        self.budget = budget
        self.needed = needed_slots
        self.w = (w_cost, w_qual, w_slot)
        self.target = budget * 0.55          # aggressive opening target
        self.accept_threshold = 0.72         # relaxes each round

    def utility(self, bid):
        """Buyer utility U = wc*(1-fee/budget) + wq*(quality/10) + ws*min(slots/needed,1)"""
        if bid["fee"] &gt; self.budget:
            return -1.0
        wc, wq, ws = self.w
        u = (wc * (1 - bid["fee"] / self.budget)
             + wq * (bid["quality"] / 10)
             + ws * min(bid["slots"] / self.needed, 1.0))
        return round(u, 4)

    def evaluate(self, bids, rnd):
        scored = [(self.utility(b), b) for b in bids if b]
        scored = [(u, b) for u, b in scored if u &gt;= 0]
        scored.sort(key=lambda x: -x[0])
        for u, b in scored:
            log(self.name, "EVALUATE", f"{b['agent']:&lt;14} buyer-utility = {u}")
        return scored
```

6. Result

The contract was awarded to MentorAgent-C in round 3 at Rs 7,036.48, with 10 slots for an 8-slot need and buyer utility 0.6661 (threshold 0.650 at that point). The two losing mentors were sent REJECT-PROPOSAL.

Execution output (log file)

Captured from **logs/3.1_negotiation.log** (python3 negotiation.py). Timestamps and some values differ between runs; this is the recorded run.

```
# Activity 3.1_negotiation
# command: python3 negotiation.py
# exit code: 0
# lines: 73


================================================================================
 MULTI-AGENT CONTRACT NET PROTOCOL :: MOCK-INTERVIEW TRAINING SLOT (Assignment 3.1)
================================================================================
[10:40:21] StudentPrepAgent --INIT          --&gt; budget=Rs.12,000 | slots needed=8 | weights cost/quality/slots=(0.45, 0.
[10:40:21] MentorAgent-A    --REGISTER      --&gt; reservation=Rs.5,200 quality=9/10 capacity=6 flexibility=0.3
[10:40:21] MentorAgent-B    --REGISTER      --&gt; reservation=Rs.3,800 quality=7/10 capacity=10 flexibility=0.45
[10:40:21] MentorAgent-C    --REGISTER      --&gt; reservation=Rs.4,600 quality=8/10 capacity=8 flexibility=0.22

================================================================================
 PHASE 1 :: CALL FOR PROPOSALS (CFP)
================================================================================
[10:40:21] StudentPrepAgent --CFP           --&gt; broadcast to 3 participants: 'need 8 mock-interview slots under Rs.12,00
[10:40:21] MentorAgent-A    --PROPOSE       --&gt; round 0: Rs.7,749.61 | 6 slots | quality 9/10
```

```
[10:40:21] MentorAgent-B    --PROPOSE       --&gt; round 0: Rs.6,599.88 | 10 slots | quality 7/10
[10:40:21] MentorAgent-C    --PROPOSE       --&gt; round 0: Rs.6,998.39 | 8 slots | quality 8/10
[10:40:21] StudentPrepAgent --EVALUATE      --&gt; MentorAgent-C  buyer-utility = 0.6676
[10:40:21] StudentPrepAgent --EVALUATE      --&gt; MentorAgent-B  buyer-utility = 0.6475
[10:40:21] StudentPrepAgent --EVALUATE      --&gt; MentorAgent-A  buyer-utility = 0.6244

================================================================================
 PHASE 2 :: MULTI-TURN PROPOSAL / COUNTER-PROPOSAL LOOP
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--- ROUND 1 ------------------------------------------------------------------
[10:40:21] StudentPrepAgent --COUNTER       --&gt; target price Rs.6,600.00 | acceptance threshold 0.720
[10:40:21] MentorAgent-A    --PROPOSE       --&gt; round 1: Rs.7,404.73 | 7 slots | quality 9/10 | own-utility=0.6468
[10:40:21] MentorAgent-B    --PROPOSE       --&gt; round 1: Rs.6,599.93 | 10 slots | quality 7/10 | own-utility=0.8158
[10:40:21] MentorAgent-C    --PROPOSE       --&gt; round 1: Rs.6,910.75 | 9 slots | quality 8/10 | own-utility=0.6891
[10:40:21] StudentPrepAgent --EVALUATE      --&gt; MentorAgent-C  buyer-utility = 0.6708
[10:40:21] StudentPrepAgent --EVALUATE      --&gt; MentorAgent-A  buyer-utility = 0.6623
[10:40:21] StudentPrepAgent --EVALUATE      --&gt; MentorAgent-B  buyer-utility = 0.6475
[10:40:21] StudentPrepAgent --CONCEDE       --&gt; no bid met threshold -&gt; relaxing to 0.685

--- ROUND 2 ------------------------------------------------------------------
[10:40:21] StudentPrepAgent --COUNTER       --&gt; target price Rs.6,996.00 | acceptance threshold 0.685
[10:40:21] MentorAgent-A    --PROPOSE       --&gt; round 2: Rs.7,282.11 | 7 slots | quality 9/10 | own-utility=0.6303
[10:40:21] MentorAgent-B    --PROPOSE       --&gt; round 2: Rs.6,778.16 | 10 slots | quality 7/10 | own-utility=0.8486
[10:40:21] MentorAgent-C    --PROPOSE       --&gt; round 2: Rs.6,929.50 | 9 slots | quality 8/10 | own-utility=0.692
[10:40:21] StudentPrepAgent --EVALUATE      --&gt; MentorAgent-C  buyer-utility = 0.6701
[10:40:21] StudentPrepAgent --EVALUATE      --&gt; MentorAgent-A  buyer-utility = 0.6669
[10:40:21] StudentPrepAgent --EVALUATE      --&gt; MentorAgent-B  buyer-utility = 0.6408
[10:40:21] StudentPrepAgent --CONCEDE       --&gt; no bid met threshold -&gt; relaxing to 0.650

--- ROUND 3 ------------------------------------------------------------------
[10:40:21] StudentPrepAgent --COUNTER       --&gt; target price Rs.7,415.76 | acceptance threshold 0.650
[10:40:21] MentorAgent-A    --PROPOSE       --&gt; round 3: Rs.7,322.20 | 7 slots | quality 9/10 | own-utility=0.6357
[10:40:21] MentorAgent-B    --PROPOSE       --&gt; round 3: Rs.7,065.08 | 11 slots | quality 7/10 | own-utility=0.9315
[10:40:21] MentorAgent-C    --PROPOSE       --&gt; round 3: Rs.7,036.48 | 10 slots | quality 8/10 | own-utility=0.7458
[10:40:21] StudentPrepAgent --EVALUATE      --&gt; MentorAgent-C  buyer-utility = 0.6661
[10:40:21] StudentPrepAgent --EVALUATE      --&gt; MentorAgent-A  buyer-utility = 0.6654
[10:40:21] StudentPrepAgent --EVALUATE      --&gt; MentorAgent-B  buyer-utility = 0.6301
[10:40:21] StudentPrepAgent --ACCEPT        --&gt; utility 0.6661 &gt;= threshold 0.650 -&gt; intent to award

================================================================================
 PHASE 3 :: CONSENSUS HANDSHAKE / CONTRACT AWARD
================================================================================
[10:40:21] StudentPrepAgent --AWARD         --&gt; CONTRACT -&gt; MentorAgent-C
[10:40:21] MentorAgent-C    --ACK           --&gt; ACCEPT-PROPOSAL received. Committing 10 slots.
[10:40:21] StudentPrepAgent --CONFIRM       --&gt; HANDSHAKE COMPLETE (3-way: CFP -&gt; PROPOSE -&gt; ACCEPT/ACK)
[10:40:21] StudentPrepAgent --REJECT        --&gt; REJECT-PROPOSAL sent to MentorAgent-A
[10:40:21] StudentPrepAgent --REJECT        --&gt; REJECT-PROPOSAL sent to MentorAgent-B

  FINAL CONTRACT
    Provider        : MentorAgent-C
    Agreed Fee      : Rs.7,036.48  (budget Rs.12,000)
    Slots Secured   : 10 / 8 required
    Trainer Quality : 8/10
    Buyer  Utility  : 0.6661
    Seller Utility  : 0.7458
    Rounds Taken    : 3
    STATUS          : CONSENSUS REACHED -- NEGOTIATION TERMINATED
================================================================================
```

7. Viva points

Why private reservation prices? Each agent protects its own limit, which makes the negotiation real rather than a fixed price.

Concession is a fixed rule here, not learned. This is a limitation to state openly.

The buyer accepts once utility passes a threshold rather than searching for the global optimum; that is a deliberate trade of optimality for speed.

8. Conclusion

Three competing sellers were brought to one agreement that satisfied the buyer's threshold, with the losers released by an explicit handshake.